# VIDYAPEETH

**BATCH CODE:** 12- AN181MA

**SUBJECT NAME:** CHEMISTRY

**CHAPTER NAME:** Atomic structure

**Lecture No.** 01

**By Amit Sir**

# Today's Goal

Subtopic .. Basic

Subtopic ..

Subtopic ..

# Chemistry

- **Physical Chemistry**
  - related with mathematical calculation
- **Inorganic Chemistry** → study of elements except C
- **Organic Chemistry** → study of carbon

## 11th Physical chemistry

1. structure of an atom (20 lec)
2. Some Basic concept of chemistry (mole concept) (15 lec)
3. Thermodynamics (15 lec)
4. Equilibrium (15 lec)
5. redox reactions (6 lec)

## 11th Inorganic chemistry

1. Periodic Table (8-10 lec)
2. chemical bonding (20 lec)
3. P-block (7-8 lec)

## 11th organic chemistry

1. Nomenclature (15 lec)
2. Isomerism (15 lec)
3. general organic chemistry (GOC) 15 lec
4. hydrocarbon 15 lec
5. Purification → 5 lec

| Element | Atomic number (Z) | Mass number (A) |
|---|---|---|
| H | 1 | 1 |
| He | 2 | 4 |
| Li | 3 | 7 |
| Be | 4 | 9 |
| B | 5 | 11 |
| C | 6 | 12 |
| N | 7 | 14 |
| O | 8 | 16 |
| F | 9 | 19 |
| Ne | 10 | 20 |

| | Z | A |
|---|---|---|
| Na | 11 | 23 |
| Mg | 12 | 24 |
| Al | 13 | 27 |
| Si | 14 | 28 |
| P | 15 | 31 |
| S | 16 | 32 |
| Cl | 17 | 35 \| 37 |
| Ar | 18 | 40 |
| K | 19 | 39 |
| Ca | 20 | 40 |

amu

Atomic mass → 35.5

# Atomic number: (Z) →

⇓

no. of protons present in an atom is called Atomic number.

$^{A}_{Z}X$

$$\boxed{Z = P}$$

Z → atomic no.

P → no. of Proton

Na

$^{23}_{11}Na$

no. of Proton = 11

## ② Mass number (A): →

⇓

summation of no. of proton and neutrons present in an atom is called Mass number.

$$\boxed{A = P + n}$$

$$n = A - P$$

$$\boxed{n = A - Z}$$

(Z = P)

n → neutron

relative charge

Atom → proton → +1
→ electron → −1
→ neutron → 0

* Atom is electrically neutral

⇓

For atom to be neutral

no. of protons = no. of electrons

P = e

* Z = P
* P = e

$\boxed{Z = P = e}$ → neutral atom

Z → atomic no
P → proton
e → electron

Ex:-

24
12 Mg

Z = 12
P = 12
e = 12
n = 24 − 12
= 12

★★★★★

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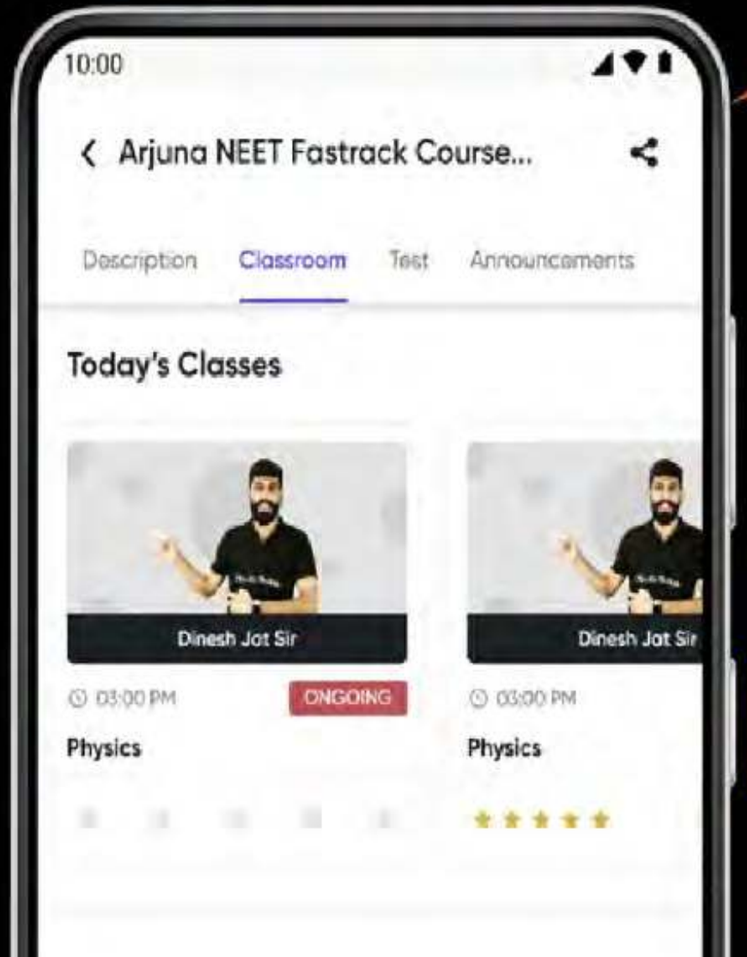
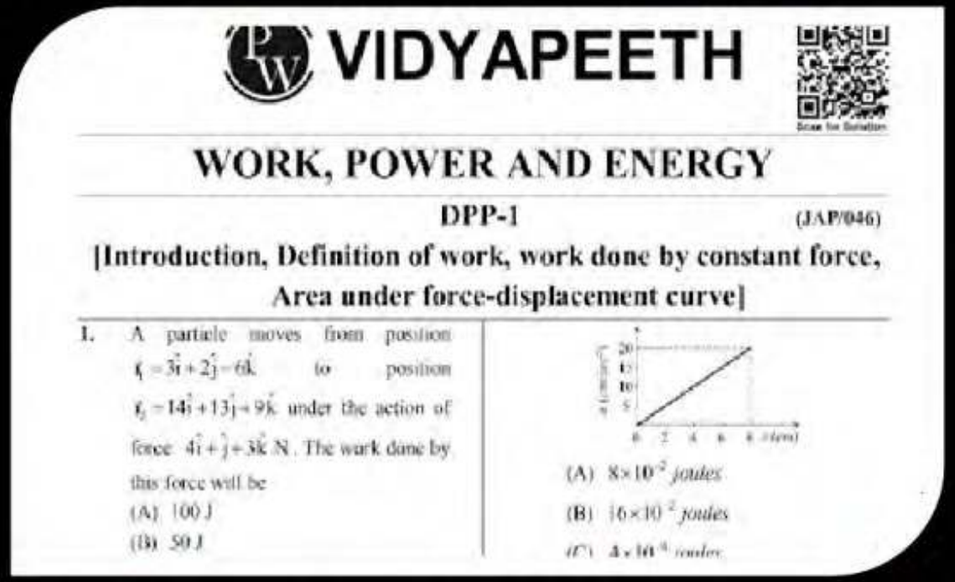
## Solve the DPP and check Solution

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